HELMET DETECTION AND NUMBER PLATE EXTRACTION

A Project Report submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

Submitted by

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DECLARATION

I/We, hereby declare that the project report entitled **HELMET DETECTION AND NUMBER PLATE EXTRACTION** is an original work done in the Department of Computer Science and Engineering, GITAM School of Technology, GITAM (Deemed to be University) submitted in partial fulfillment of the requirements for the award of the degree of B.Tech. in Computer Science and Engineering. The work has not been submitted to any other college or University for the award of any degree or diploma.

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This is to certify that the project report entitled **HELMET DETECTION AND NUMBER PLATE EXTRACTION** is a bonafide record of work carried out by **KASINADHUNI PRANAV SAI(222010305025),GUNDLAPALLY VENKATA SAI SESHAGIRI RAO(222010306022),NADIMPALLI VENKATA SAI SAMARTH(222010306023),KOLLA HANISH CHANDRA(222010306036)** students submitted in partial fulfillment of requirement for the award of degree of Bachelors of Technology in Computer Science and Engineering during the academic year 2023- 2024.

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LIST OF SYMBOLS AND ABBREVIATIONS

CSV	-	Comma Separated Values
CNN	-	Convolutional Neural Network
R-CNN	-	Region-Based Convolutional
RPN	-	Region Proposal Network
RNN	-	Recurrent Neural Network
UML	-	Unified Modelling Language
YOLO	-	You only look once
MVD	-	Motor vehicle dealer/department
HOG	-	Histogram of oriented gradient
LBP	-	Local Binary Pattern
SVM	-	Support Vector Machine
MLP	-	Multilayer Perceptron
RBFN	-	Radial Basics Function Network
VGG/VGG16/VGG19	-	Visual Geometry Group
ROI	-	Region of Interest
CUDA	-	Compute unified device architecture
GPU	-	Graphics processing unit

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1. Introduction

In recent years, there has been a growing concern for road safety, particularly regarding two critical aspects: helmet usage by riders and proper vehicle identification through number plates. To address these concerns, advanced technologies in computer vision and image processing have been employed. This project aims to develop a system for detecting helmets worn by riders and extracting number plates from vehicles using machine learning algorithms.

1.1 Problem Definition

The primary problem addressed in this project is the need for enhancing road safety through effective enforcement of helmet-wearing regulations and ensuring proper vehicle identification. Currently, manual enforcement of these regulations is time-consuming and inefficient. Moreover, human errors can lead to inaccuracies and inconsistencies in the enforcement process.

1.2 Objective

The main objective of this project is to develop an automated system capable of:

- Detecting whether riders are wearing helmets or not.
- Extracting and recognizing vehicle number plates accurately.

1.3 Limitations

While this project aims to provide an automated solution for helmet detection and number plate extraction, it is essential to acknowledge certain limitations:

- **Environmental Factors:** The performance of the system may be affected by varying lighting conditions, weather conditions, and occlusions.
- **Camera Quality:** The accuracy of object detection and recognition heavily relies on the quality of the input images captured by the camera.
- **Computational Resources:** The system's performance may be constrained by the computational resources available, particularly for real-time applications.

1.4 Outcomes

Upon the successful completion of this project, the following outcomes are expected:

- Development of a robust helmet detection algorithm capable of accurately identifying whether riders are wearing helmets.
- Implementation of a number plate extraction system that can accurately locate and recognize vehicle number plates.
- Integration of the above algorithms into a cohesive software application with a user-friendly interface.
- Evaluation of the system's performance through rigorous testing and validation processes.

1.5 Applications

The proposed system has various applications, including:

- **Traffic Enforcement:** Law enforcement agencies can utilize this system to enforce helmet-wearing regulations and monitor vehicle registration compliance effectively.
- **Road Safety Awareness:** By automatically detecting helmet usage and vehicle identification, the system can contribute to raising awareness about road safety among riders and drivers.
- **Surveillance Systems:** The technology developed in this project can be integrated into surveillance systems for monitoring traffic flow and identifying vehicles involved in incidents or criminal activities.

In conclusion, the development of an automated system for helmet detection and number plate extraction holds significant potential for improving road safety and enhancing law enforcement capabilities. By leveraging advancements in computer vision and machine learning, this project aims to provide an effective and efficient solution to address these critical issues.

2. Literature Review

Different approaches have been put forth by numerous researchers to address the issue of autonomous helmet identification in real-time traffic situations. The section below discusses these techniques.
In the beginning, researchers employed computer vision techniques like HOG, SURF, and SIFT for machine learning to automatically detect whether motorcycle riders were wearing helmets or not.
In recent years, several studies were performed to analyze traffic on public roads, including the detection, classification and counting of vehicles and helmet detection. The detection and segmentation of vehicles on public roads can be considered as the first step to develop any study related to vehicular traffic. For this reason, some relevant studies are discussed in this section.

Kunal Dahiya[1] proposed a framework for real-time detection of traffic rule violators who ride bikes without using a helmet. Proposed framework will also assist the traffic police for detecting such violators in odd environmental conditions viz; hot sun, etc. Experimental results demonstrate the accuracy of 93.80% for detection of violators. Average time taken to process a frame is 11.58 ms, which is suitable for real time use. Also, the proposed framework automatically adapts to new scenarios if required, with slight tuning. This framework can be extended to detect and report number plates of violators.

A computational vision system was proposed by Chiu C-C. [2] with the aim of recognizing and tracking motorcyclists that are partially obscured by another vehicle. A system for counting vehicles is suggested. A motorbike is identified using a helmet detecting system. The system assumes the area around the helmet has a form resembling a circle. The boundaries of the image are estimated across the helmet's potential zone, which is the area surrounding the motorcycle, in order to identify the helmet. Then iteratively count the number of edge points that approximate a circle. During system calibration, the region will correlate to a helmet if this number is more than or equal to a present value. When a helmet is found, the system assumes that a motorbike is nearby. The system operator must enter some parameters during its calibration stage, including the helmet radius, camera angle, and height. All settings should be changed if any condition, such as a change in camera height or the road where the device is being used, occurs.

A solution that relies on traffic engineering techniques to detect moving cars was developed by Leelasantitham and Wongseree [3]. There were five sets of automobiles. Bicycles, motorcycles, and tricycles made up the first group. Vehicles in the second group included vans and sedans. The third category includes small-sized vehicles and minibusses. Large buses and medium-sized vehicles made up the fourth group. Large vehicles and trailers made up the sixth group. The image's length and breadth were the features that were used for classification. Only 76 photos make up the database, which calls into question the validity of the findings. Keep in mind that the system is very dependent on the road on which the photographs are taken; therefore, all parameters, such as the length and breadth of the image, should be changed if the same system is applied to a different road.

Tahniyath Wajeeh[4] proposed a method using image processing to provide information on the population of traffic violators in a certain area. It creates a database of all bikers who drive without a helmet and takes a photo as evidence.
The program is considerably less expensive thanks to the use of open-source, free technologies like tesseract, Open CV, and tensor flow. This technology was tested to provide complete proof and accurate findings under good lighting circumstances. The impact of the system will rise as more people become aware of it.

Messelodi S. [5] presented a segmentation and classification method for vehicles. To categorize the artifacts, eight three-dimensional models were made. The size of the vehicle, which changes depending on the type of vehicle, was used to calculate these models. Cycle (motorcycles and bicycles), small automobile, car, minibus, closed truck, open truck, bus, and pedestrian were among the models that were built. Each captured vehicle has a model that is generated, and these models are compared to the models made for each class of vehicle. The vehicle class is determined by the model that is most similar to the model of the captured vehicle. This study's use of a single model for both bicycles and motorcycles has a drawback. Furthermore, it has been established that using solely geometry data to categorize the vehicles is insufficient for describing these kinds of objects. The fact that some parameters like camera height, angle, and focal distance should always have the same values is another drawback of this approach. No vehicle would fit the models if these characteristics were altered, hence new ones should be made. The system could not be used on some public highways because they lacked a site where the camera could be positioned, as stated in this paper.

Rohith C A[6] proposed a system for an Efficient Helmet Detection for MVD using Deep Learning.The condition selected for classifying the captured frames is a person wearing a helmet and non-helmet while driving a two-wheeler. Firstly, they classify images that contain a person traveling in a two-wheeler or not. Then, analyze the classified images to obtain the data set based on the condition. The captured frames that contain images other than the above mentioned condition will be discarded or not considered. The model used for the detection and extraction is Caffe model. The proposed model shows an accuracy score of 86%. The model used for classification is the Inception V3 model. The proposed model showed a validation accuracy score of 74%.

Prity Kumari[7] proposed a system which is extremely effective for the protection purpose of the user. Users need to wear a helmet to ride a motorcycle and therefore traffic rules are monitored by the rider. This method is below pocket management that's riding the two wheeler vehicle having safety in hand and is affordable. This method consumes simple functionalities. It provides an improved security to the biker.
All the libraries and software system employed in project are open supply and henceforward is extremely versatile and budget good. The project was primarily engineered to unravel the matter of non-efficient traffic organization. Henceforward at the tip of it will say that if organized by any traffic management departments, it'd create their job easier and additional economical.

Romuere Silva[8] showed that the LBP descriptor proved to be more robust for the problem than HOG and Haar Wavelet descriptors. The LBP descriptor describes the local texture structure by pattern joint distribution. The texture patterns in motorcycles contribute to good performance of the classifier using the LBP features. SVM classifiers presented good results. The main advantage of the SVM is the performance in the training phase. The SVM is usually faster in the training process than Multi-layer Perceptron (MLP) and Radial Basis Function Network (RBFN).

In the helmet detection step the Random Forest algorithm obtained the best result. The combination between CHT, HOG and LBP return a good satisfactory result in helmet detection. This can be explained by combining information edge (HOG), texture (LBP) and geometric (CHT) information to build the feature vector.

GayathriV[9] study recommends a system for monitoring motorcycle riders who don't wear helmets and defaulters who ride three motorcycles at once. A PC vision framework with modules for moving item division, moving item arrangement, and helmet use identification will help the traffic experts. The suggested framework will also help the traffic police track down such criminals in difficult conditions, such as scorching heat, etc. Propelled following computations are frequently required to deal with obstacles. When there is no light, cameras with night vision are frequently used to use the location framework. In the future, more instances of both positive and negative behavior will be remembered in response to requests to improve the framework's capacity for conjecture. Additionally, work with the front-end video catch modules.

The research by Narong Boonsirisumpun[10] described an experiment that is conducted to address the problems of motorcycle riders wearing a helmet and not wearing a helmet detection and categorization. In these tests four convolutional neural networks (CNNs) (VGG16, VGG19, Google Net or Inception V3, and Mobile Nets) are used, and combine these models with the SSD approach to perform an image detection step.

Romuere Silva[11] discusses how to identify motorcycle riders who are not wearing helmets. They suggest a computer vision system that is separated into three parts: Segmentation of moving objects, classification of moving objects, and detection. The outcomes meet expectations. The best outcome was obtained by the MLP classifier employing the HOG descriptor, which had an Accuracy of 0.9137.

The Lucas-Kanade tracker algorithm [12] was employed as the tracking algorithm. The tracking of objects determines whether a vehicle and a person will collide. If a pedestrian and a vehicle simultaneously traverse the intersection, the system will issue a warning. The possibility of false positives in this system is high as a pedestrian can change routes after the system issues the warning.

Dr. A. Radhika[13] suggested The convolutional neural network-based solution uses two-wheeler and motorcycle riders to detect helmet use. The model employs the YOLO algorithm with SSD Mobile Net to identify bike riders wearing safety helmets. To train and test the model, a collection of images featuring different helmets is created and divided into three sections. The model is trained using the TensorFlow framework. After training and testing, the detection model's mean average precision (mAP) is stable, and a helmet detection model is created.

C. Vishnu[14] suggested framework for the automatic detection of motorcycle riders operating without helmets uses adaptive background subtraction, which is resistant to a variety of issues including lighting, bad video quality, etc. Making advantage of the detection rate is increased and the number of false alarms is decreased by using deep learning for automatic learning of discriminative representations for classification tasks, leading to a more trustworthy system. On two genuine video datasets, the experiments effectively identified 92.87% of violators with a low false alarm rate of 0.50%, demonstrating the effectiveness of the suggested method.

Jimit Mistry[15] provides a CNN-based method for automatic helmet detection. They employ two stage YOLOv2 models to improve the accuracy of helmet detection. The YOLOv2 model, which was trained on the COCO dataset, first assures person detection. By taking this action, fewer helmets will go undetected. The second YOLOv2 model, which was trained using our data set of helmeted photographs, receives input from the cropped images of the observed person. Various helmeted image scenarios have been used to test the suggested methodology. Additionally, they assessed the quantitative measures on the test photos and contrasted them with other cutting-edge techniques. Experimental findings and quantitative measurements on various scenarios demonstrate the robustness and dependability of our technique, which has a high helmet detection accuracy.

3. Problem Analysis

3.1 Problem Statement

The project aims to address the critical issues of road safety and law enforcement by developing a system capable of detecting helmet usage among riders and extracting number plates from vehicles.

3.2 Existing System

Currently, the enforcement of helmet-wearing regulations and vehicle identification relies heavily on manual intervention by law enforcement officers. This manual approach is not only time-consuming but also prone to human errors, leading to inconsistencies and inaccuracies in enforcement. Additionally, traditional methods do not provide real-time monitoring capabilities, limiting their effectiveness in ensuring road safety.

3.3 Flaws & Disadvantages

The flaws and disadvantages of the existing system include:

- **Inefficiency:** Manual enforcement is labor-intensive and time-consuming.
- **Inaccuracy:** Human errors may lead to incorrect judgments and enforcement.
- **Lack of Real-time Monitoring:** Traditional methods do not offer real-time monitoring capabilities, making it challenging to enforce regulations effectively.
- **Limited Coverage:** Manual enforcement efforts are limited in scope and cannot monitor all vehicles simultaneously.

3.4 Proposed System

The proposed system aims to overcome the limitations of the existing system by employing advanced technologies in computer vision and machine learning. By utilizing image processing techniques, the system will detect whether riders are wearing helmets and extract vehicle number plates automatically. This automated approach will enable real-time monitoring and enforcement of road safety regulations, significantly enhancing the effectiveness of law enforcement efforts.

3.5 Functional Requirements

The functional requirements of the proposed system include:

- **Helmet Detection:** The system should be able to detect whether riders are wearing helmets or not.
- **Number Plate Extraction:** The system should accurately extract vehicle number plates from images or video feeds.
- **Real-time Monitoring:** The system should provide real-time monitoring capabilities to enable immediate enforcement actions.
- **Accuracy:** The system should deliver high accuracy in helmet detection and number plate extraction to minimize false positives and negatives.
- **Scalability:** The system should be scalable to accommodate varying traffic volumes and environments.

3.6 Non-Functional Requirements

The non-functional requirements of the proposed system include:

- **Performance:** The system should perform efficiently, with minimal latency in processing.
- **Reliability:** The system should be reliable and robust, capable of functioning under different environmental conditions.
- **Usability:** The system should have a user-friendly interface, allowing law enforcement personnel to operate it easily.
- **Security:** The system should ensure the security and privacy of captured data and sensitive information.
- **Scalability:** The system should be scalable to accommodate future enhancements and updates.

In conclusion, the proposed system for helmet detection and number plate extraction offers a comprehensive solution to enhance road safety and improve law enforcement effectiveness. By leveraging advanced technologies, the system aims to overcome the limitations of the existing manual enforcement methods and provide a more efficient and accurate approach to monitoring and enforcing regulations.

4. System Design

4.1 Proposed System Architecture

4.1.1 main.py

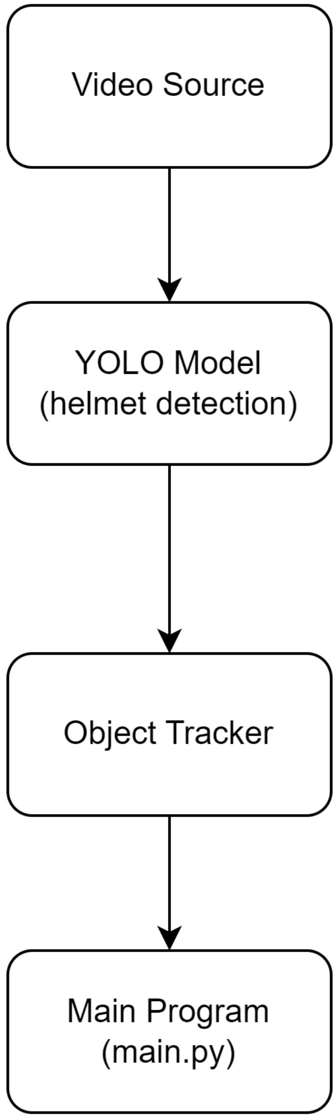

Figure 4.1.1

4.1.2 speed.py

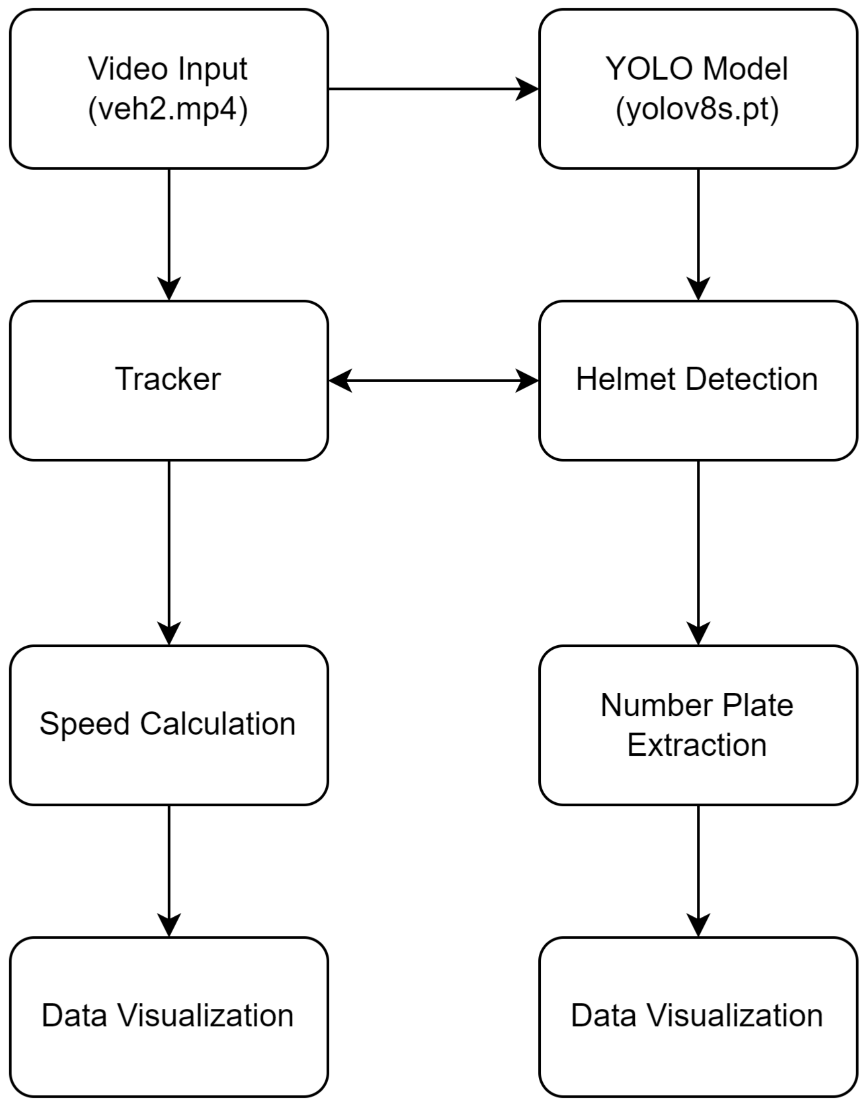

Figure 4.1.2

4.1.3 tracker.py

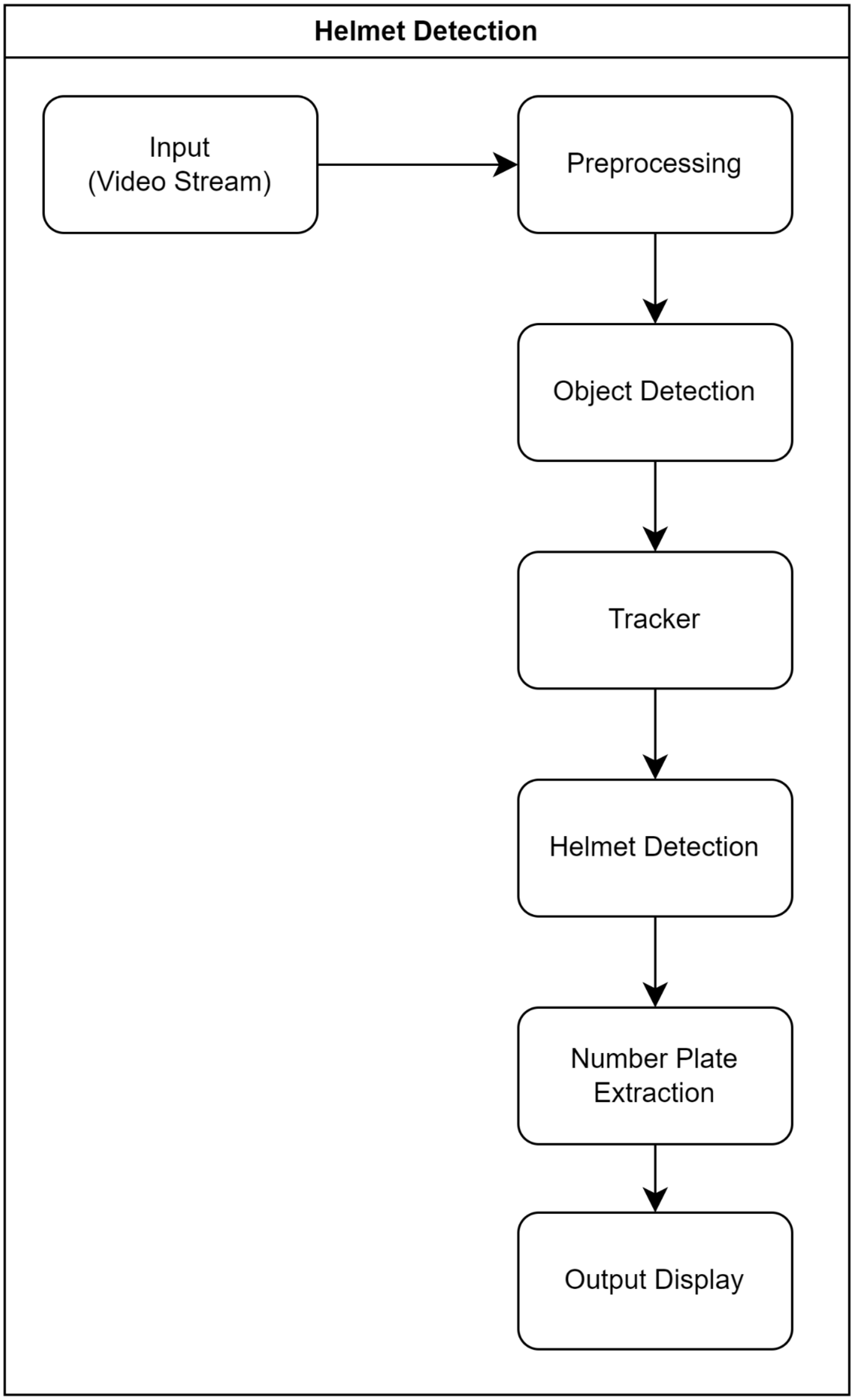

Figure 4.1.3

4.2 UML Diagrams
4.2.1 Use Case Diagram

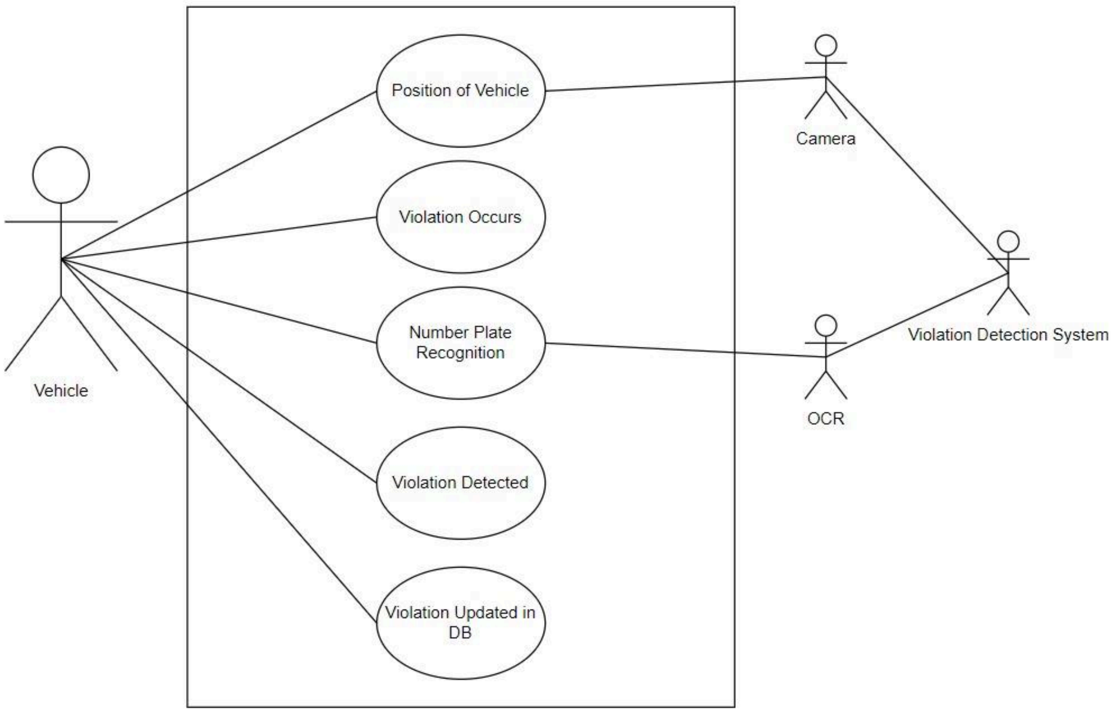

Figure 4.2.1

4.2.2 Sequence Diagram

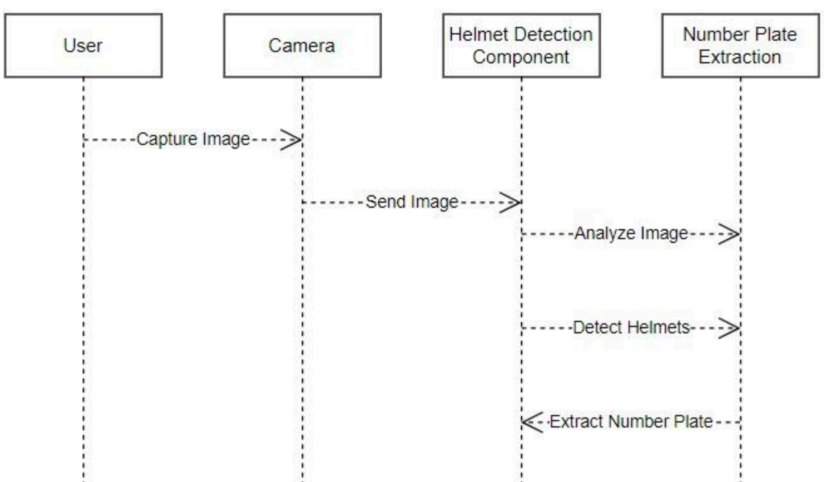

Figure 4.2.2

4.3 MVC Diagram

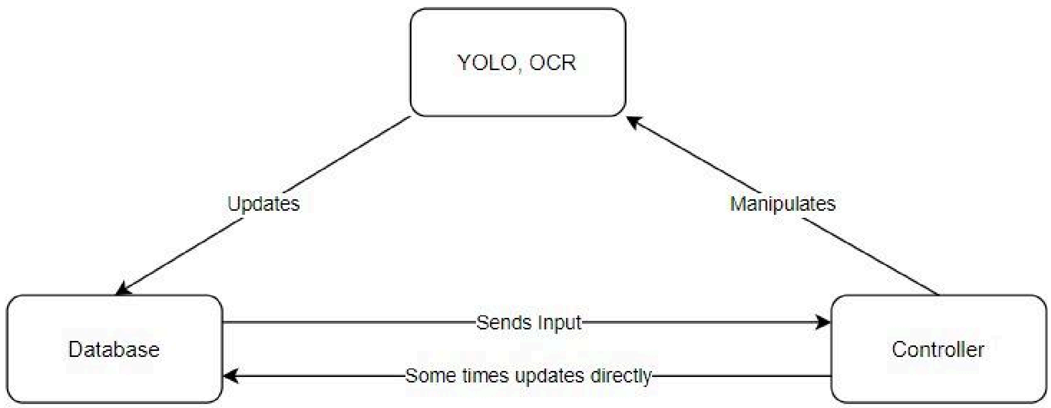

Figure 4.3

5. Implementation

5.1 Overview of Technologies

In this section, we provide an overview of the technologies used in the implementation of the project.

5.1.1 Python 3

Python 3 was chosen as the primary programming language for its simplicity, versatility, and extensive library support. Its readability and ease of use make it suitable for rapid development of computer vision applications.

5.1.2 Windows OS

Windows operating system was selected as the development environment due to its widespread use and compatibility with a wide range of software tools and libraries commonly used in machine learning and computer vision projects.

5.1.3 Visual Studio

Visual Studio, an integrated development environment (IDE), was utilized for code development and debugging. Its user-friendly interface and comprehensive set of features streamline the development process.

5.1.4 MP4

MP4 is a popular file format for storing multimedia content, including video files. It was chosen as the input format for video data used in the project due to its widespread support and compatibility with OpenCV.

5.1.5 OpenCV

OpenCV (Open Source Computer Vision Library) is a powerful open-source library for computer vision and image processing tasks. It provides various functions and algorithms for tasks such as object detection, image manipulation, and feature extraction, making it an essential component of the project.

5.2 Libraries Imported

In this subsection, we list and briefly describe the libraries imported and utilized in the project.

5.2.1 Pandas

Pandas is a Python library commonly used for data manipulation and analysis. It was used in the project for handling and processing tabular data related to object detection results and license plate information.

5.2.2 Ultralytics

Ultralytics is a deep learning library that provides pre-trained models and utilities for object detection tasks. It was used in conjunction with YOLO (You Only Look Once) for helmet detection and object tracking.

5.2.3 CV2 (OpenCV)

CV2, the Python interface to OpenCV, was extensively used for various image processing tasks such as reading video files, image manipulation, object detection, and drawing bounding boxes.

5.2.4 YOLO

YOLO (You Only Look Once) is a popular real-time object detection algorithm known for its speed and accuracy. It was utilized for detecting helmets and other objects of interest in the video frames.

5.2.5 Numpy

Numpy is a fundamental library for numerical computing in Python. It was used for efficient array manipulation and mathematical operations, particularly in the context of image processing and data handling.

5.2.6 Time

The Time module in Python was used for tracking and measuring the execution time of various operations and algorithms, aiding in performance optimization and debugging.

5.2.7 Math

The Math module provides mathematical functions and constants in Python. It was used for various mathematical calculations required in the project, such as angle calculations and geometric transformations.

5.2.8 Tracker

Tracker is a library used for object tracking in videos. It enables the tracking of objects across consecutive frames, facilitating tasks such as monitoring the movement of detected objects.

5.3 Dataset

Acquiring a well-curated dataset is essential for training and evaluating the effectiveness of the proposed system for helmet detection and number plate extraction for two-wheelers. This dataset should encompass various scenarios and conditions encountered in real-world environments to ensure the robustness and accuracy of the developed algorithms.

5.3.1 Detect people who are driving a motorcycle

The dataset should include a diverse collection of images or videos depicting individuals riding motorcycles. These data should cover different environments, including urban streets, highways, and rural roads, and should capture variations in lighting, weather conditions, and traffic density. Annotated data indicating the presence of a motorcycle rider within each image or frame is crucial for training the detection algorithm.

Link:
https://www.kaggle.com/datasets/savanagrawal/detect-person-on-motorbike-or-scooter

5.3.2 Detect whether the driver is wearing a helmet or not

For training the helmet detection algorithm, annotated images or videos indicating the helmet-wearing status of the rider are necessary. The dataset should contain instances of riders wearing helmets correctly, wearing helmets incorrectly (e.g., improperly fastened), and not wearing helmets at all. Accurate annotations marking the location of the helmet within each image or frame are essential for training the algorithm effectively.

Link: https://universe.roboflow.com/bike-helmets/bike-helmet-detection-2vdjo/dataset/1

5.3.3 Detect the license plate and extract the registration number

To train the number plate extraction algorithm, a dataset comprising images or videos of two-wheelers with visible license plates is required. Annotations should precisely delineate the boundaries of the license plate within each image or frame and extract the corresponding registration number. The dataset should include various types of two-wheelers, such as motorcycles and scooters, to ensure the algorithm's robustness across different vehicle categories.

Link:
https://universe.roboflow.com/augmented-startups/vehicle-registration-plates-trudk/dataset/1

6. TESTING & VALIDATION

6.1.main.py

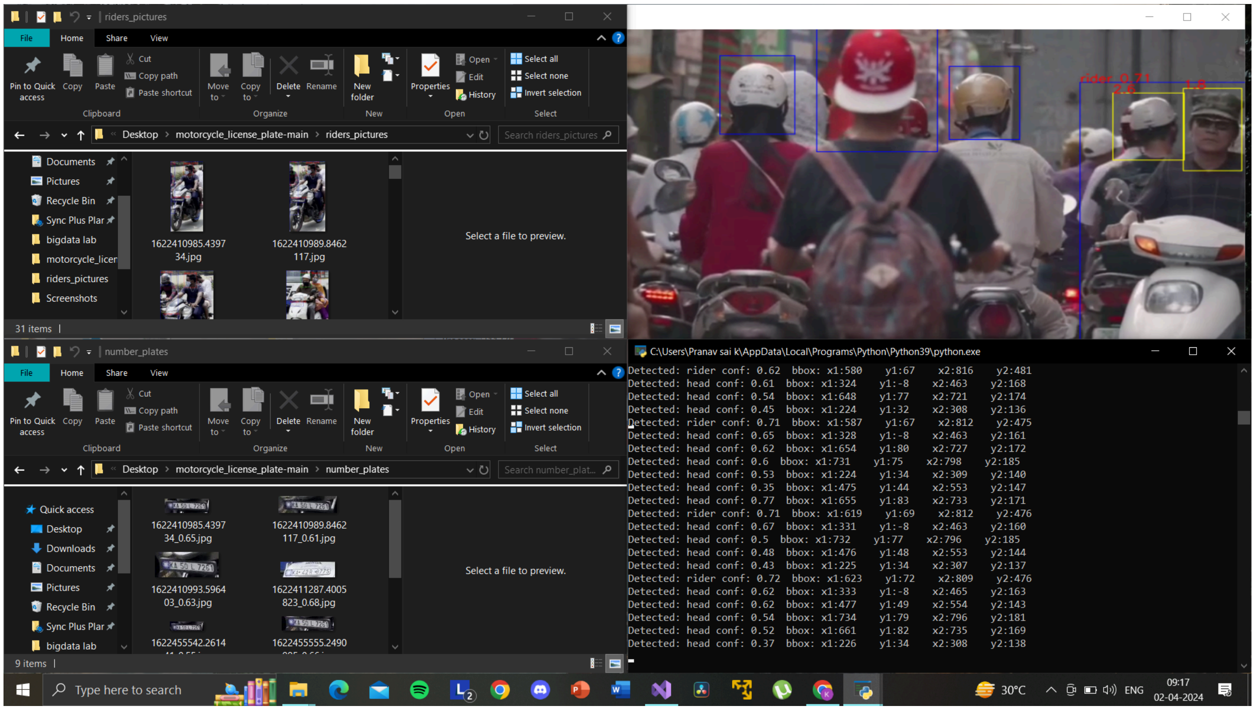

Figure 6.1

6.2.speed.py

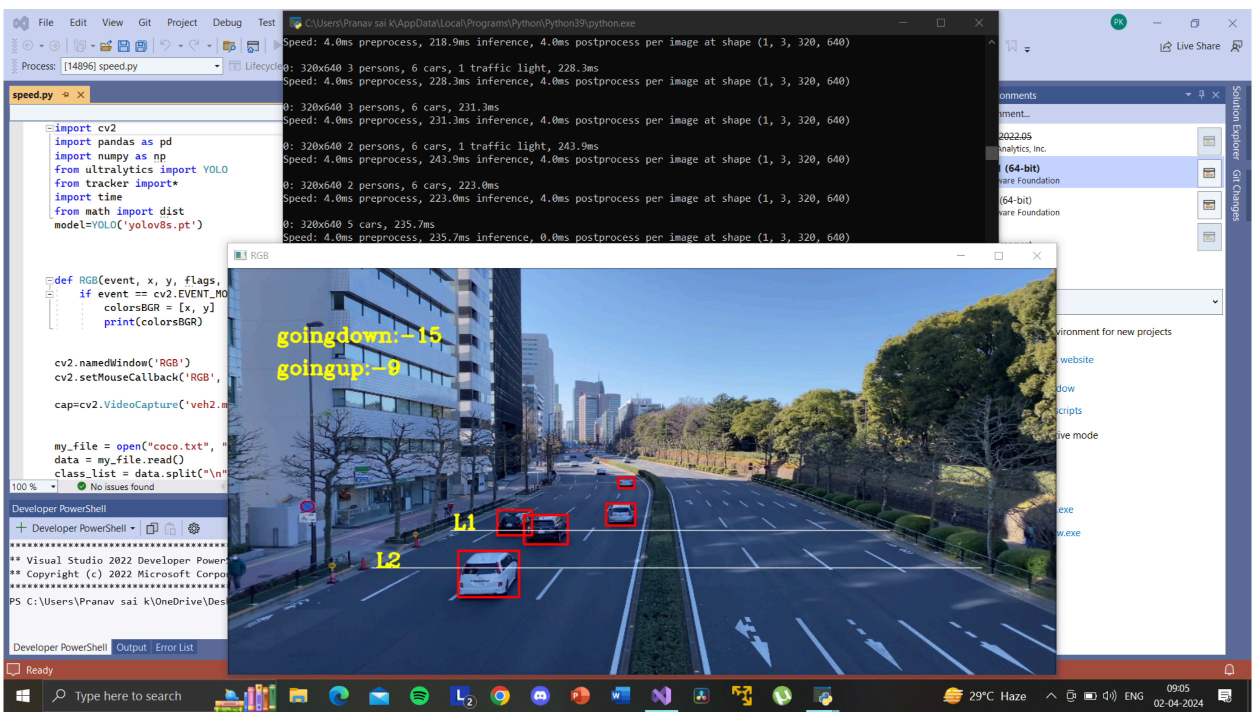

Figure 6.2

6.3.tracker.py

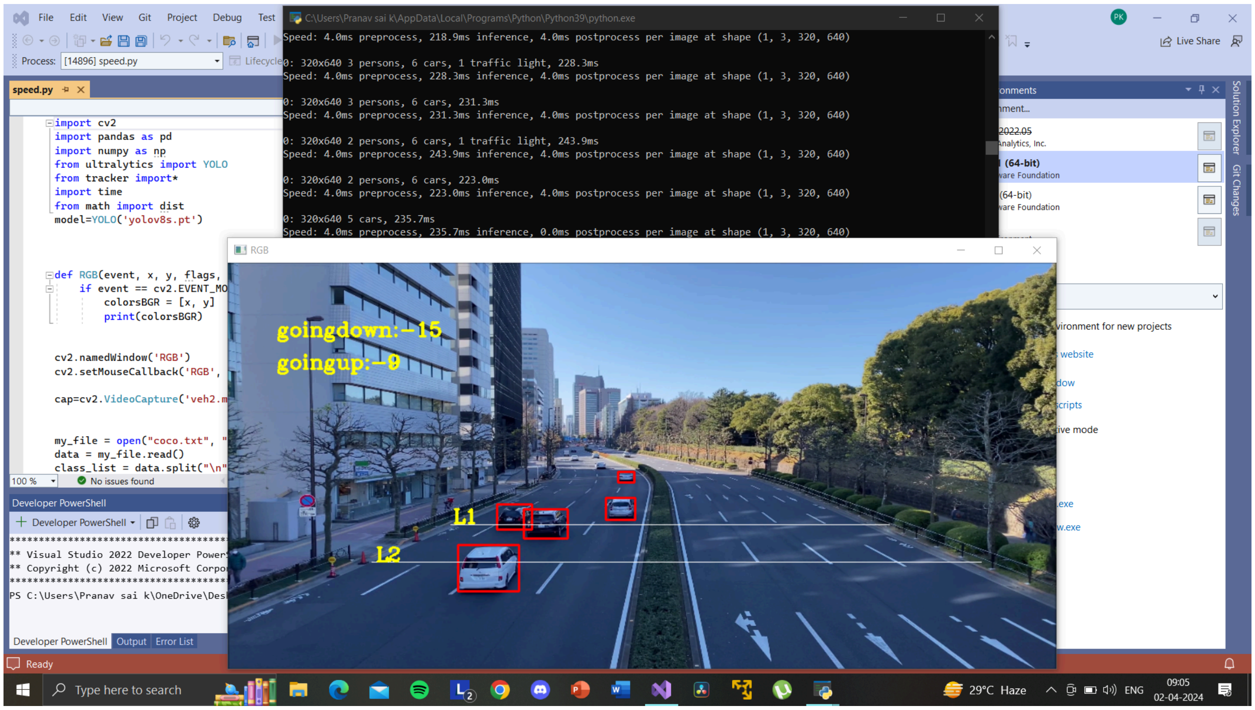

Figure 6.3

7. Result Analysis

The analysis showcases the performance evaluation of YOLOv8 models for detecting various elements relevant to road safety and law enforcement, namely, individuals driving motorcycles, helmet detection, and number plate extraction.

Motorcycle with Person Detection:

The YOLOv8 model demonstrates a notable precision of 70.4% at its best checkpoint. However, considering the importance of a balance between precision and recall, the model chosen at its best checkpoint yielded a precision of 47.5% and a recall of 43%. Further improvement is anticipated through the augmentation of the dataset size and increasing the number of epochs. Testing images reveal the model's efficacy, particularly when the motorcycle with a person is prominently visible. Augmenting the dataset with diverse images from varied angles and distances is proposed to enhance the model's performance further.

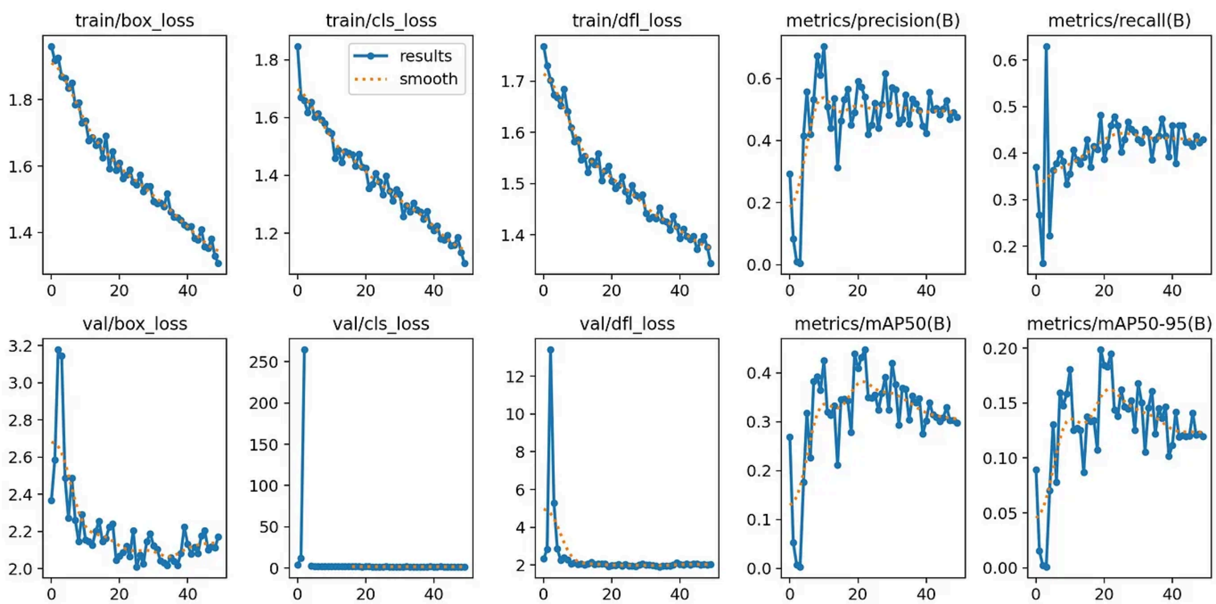

Figure 7.1

Helmet Detection:

The helmet detection model exhibited a commendable precision of 76.4% and a recall of 71.4%, with a mean average precision of 77.2%. Similar to previous analyses, potential improvements can be realized by expanding the dataset size and augmenting the training epochs. Noteworthy is the model's resilience in providing accurate results even with unclear images, which can be attributed to the extensive training dataset comprising over 1.3K images across 100 epochs.

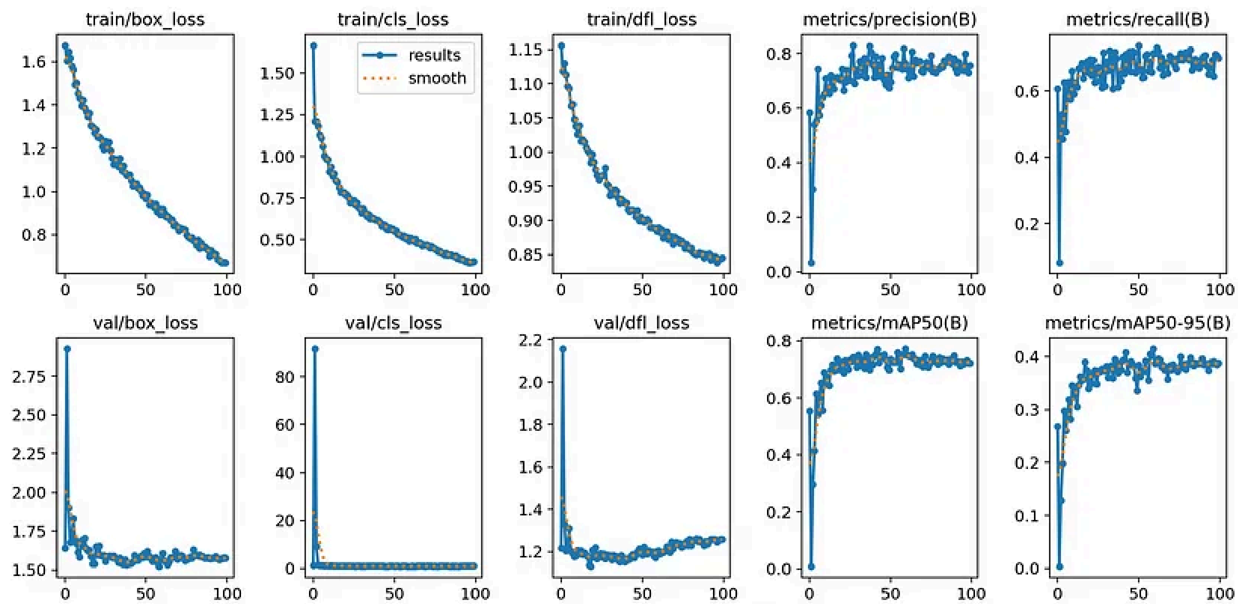

Figure 7.2

Number Plate Extraction:

The evaluation of the YOLOv8 model for number plate extraction reveals promising results. The model exhibits proficiency in accurately predicting the location of number plates, even in scenarios where the images lack clarity. The model's performance is substantiated by the accuracy, recall, and loss details presented in the training file. Testing images affirm the model's effectiveness in delineating number plate boundaries, indicative of its potential utility in license plate recognition applications.

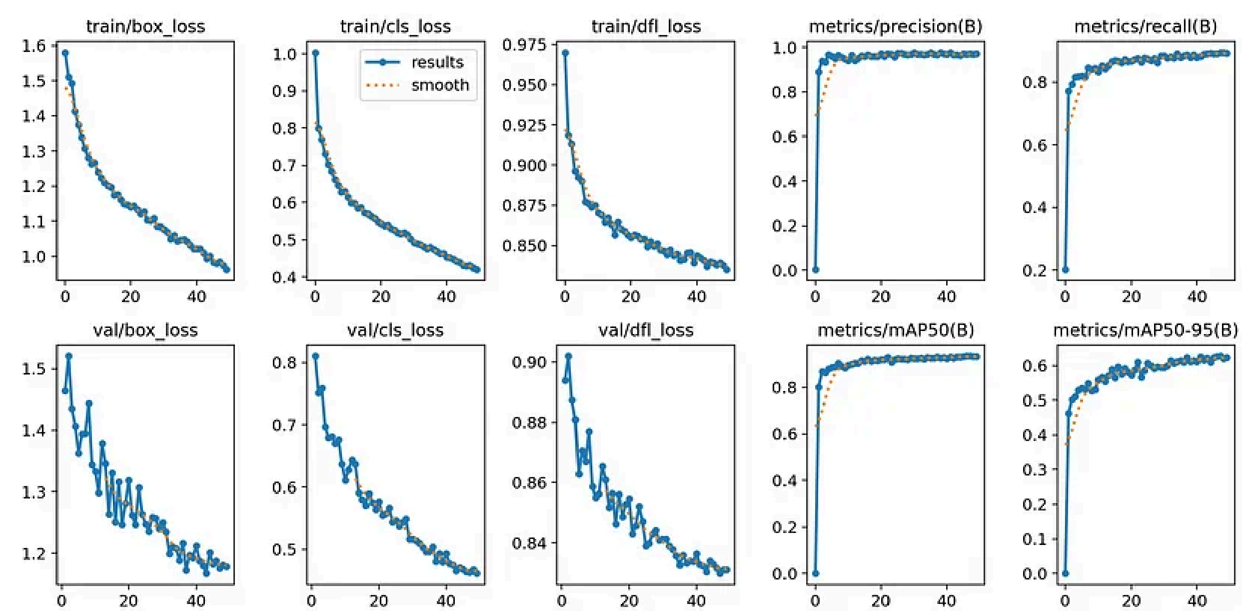

Figure 7.3

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